Isha Kulkarni

Southampton, UK • 07553993694 • ishakulkarnimscs@gmail.com • LinkedIn • github.com/ishakulkarni011

Summary

Software engineer with 2+ years building and running production software end to end in Python, TypeScript, React and Java, with REST APIs, Docker and CI/CD across AWS, Azure and GCP at 99.9 percent uptime. Automated QA and release work to cut manual workload by 60 percent, and uses Claude Code and GitHub Copilot daily with careful review and testing. MSc Computer Science, Southampton.

Technical Skills

Languages: Python • Java • TypeScript • JavaScript • C++ • SQL • HTML • CSS
Backend & Frontend: FastAPI • Flask • REST APIs • Microservices • React • Node.js • MySQL • MongoDB
Engineering Practice: Data Structures • Algorithms • OOP • Automated Testing • Code Review • Debugging • Git
Cloud & DevOps: AWS • Azure • GCP • Docker • CI/CD • Linux • Monitoring • Observability
AI & Tools: LLMs • RAG • AI Agents • Vector Databases • Claude Code • GitHub Copilot • PyTorch

Experience

Persistent Systems Limited Aug 2023 – July 2025
Software Engineer • Full Stack Development, Cloud Services, AI Applications *Pune, India*

- Built LLM powered applications and machine learning features with LangChain, retrieval augmented generation (RAG), and embeddings, through prototyping and experimentation that cut manual intervention by 70 percent.
- Engineered FastAPI microservice architectures and REST APIs with Docker containers and CI/CD pipelines on AWS, Azure, and GCP cloud infrastructure, sustaining 99.9 percent uptime and under 500ms latency with logging, monitoring, and observability.
- Enhanced quality through model evaluation and debugging before release, and delivered 50+ workshops mentoring engineers with clear communication, improving accuracy by 50 percent.
- Designed, built, and shipped production applications end to end across 10+ enterprise accounts, owning requirements, implementation, accountability, and release.
- Collaborated with product managers and business stakeholders to understand customer problems, translate requirements into scoped solutions, and contribute ideas, automating workflows that cut manual QA workload by 60 percent.

Revyamp Consultancy Oct 2022 – Jun 2023
Software Engineer Intern *Pune, India*

- Implemented a Python OCR pipeline with Tesseract and OpenCV that processed 500+ invoices daily into structured records, cutting data entry time by 60 percent.
- Developed a React.js and Node.js dashboard for real-time pipeline monitoring, giving client teams live status and cutting query resolution time by 40 percent.

Projects

Crawl AI, Website to RAG Knowledge System | *Python, LangChain, FastAPI, ChromaDB* GitHub

- Built a Python crawler that indexed 100+ pages into a ChromaDB vector database, cutting search time by 40 percent.
- Deployed a containerised FastAPI service with LangChain retrieval chains, reaching 95 percent answer accuracy on domain queries and returning answers with the source pages they came from.

Test Script Migrator, LLM Test Migration Tool | *Python, FastAPI, Streamlit, Local LLM* GitHub

- Developed a framework with AST aware parsing to migrate legacy Protractor tests to Selenium using an LLM with prompt chaining, handling 90 percent of migration patterns automatically.
- Designed maintainable, testable modules separating parsing, inference, and code generation, with automated testing and documentation that cut migration errors by 75 percent across 500+ test scripts.

Query Optimiser, Java Database Engine | *Java, SQL, Algorithms* GitHub

- Implemented 2 modules in Java with object-oriented design, a cost estimator and a greedy left-deep join optimiser, calculating efficient SQL query plans for a relational database engine.
- Applied iterative development and careful design trade-offs to compare join orders and estimate costs, with clear, testable classes for each stage.

Education & Certifications

University of Southampton Sep 2025 – Sep 2026
Master of Science in Computer Science *Southampton, UK*

- Completed modules in Machine Learning, NLP, and Deep Learning, building understanding of machine learning fundamentals and the AI development lifecycle, alongside 5+ projects on GitHub as open source contributions; MSc dissertation released a reproducible multimodal machine learning pipeline with 100+ scripted runs.

Savitribai Phule Pune University Aug 2019 – Jul 2023
Bachelor of Engineering in Information Technology *Pune, India*

- Achieved 9.02 / 10.0 CGPA in the top 3 percentile of the cohort, with coursework in programming, data structures, algorithms, databases and software engineering, while completing a software engineering internship in parallel.

AWS Certified Cloud Practitioner | *Amazon Web Services* May 2024 – May 2027

Work Authorisation: Graduate UK visa route Dec 2026 (no sponsorship required)